

GoodWhale: Helping people invest confidently with personalized, AI-driven financial literacy

Tell us your challenge. We're here to help.

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ABOUT GOODWHALE



About GoodWhale

To democratize financial literacy and align values to their personal finance, GoodWhale uses Gemini to equip people with investment knowledge through content and interactive courses.

GoodWhale is a personalized finance education platform to help aspiring investors kickstart and achieve good and enduring investing health. It aims to make money intuitive, personalized and incredibly simple with AI-generated quests, insights, stock analysis and modules.

Industries: Financial Services

Location: Singapore

Google Cloud results

- Reduces load time by 200% with Cloud Memorystore
- Out-of-the-box solutions provide simplicity and flexibility to operate without a DevOps team
- Supports better planning and forecasting with transparent pricing mode
- Ability to operate on a large scale with a lean team, and scale down quickly post-event
- Eases onboarding of new employees

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About CloudMile

CloudMile is a cloud and AI company and Google Cloud partner, empowering businesses to accelerate digital transformation through cloud technology and machine learning. With a mission to make digital changes possible for every enterprise by modernizing and integrating mainstream IT and deploying digital solutions at scale, CloudMile helps enterprises maximize the efficiency of the cloud by combining enterprise data with cloud architecture design.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Build (<https://cloud.google.com/build/>)

Cloud Memorystore
(<https://cloud.google.com/memorystore/>)

Cloud Functions
(<https://cloud.google.com/functions/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Datastore
(<https://cloud.google.com/datastore/docs/concepts/overview>)

enterprises maximize the efficiency of the

with a
familiar
cloud
infrastructure

Cloud Firestore
(<https://cloud.google.com/firestore/>)

Looker Studio
(<https://cloud.google.com/looker-studio>)

Secret Manager
(<https://cloud.google.com/secret-manager>)

Technology and digitalization has changed the way we live. This includes the way we manage our finances. With so many avenues to invest in, people have more access to financial services platforms than ever before. However, often, the lack of knowledge and financial literacy prevents people from making the most of the opportunities provided to them.

GoodWhale (<https://goodwhale.com/>), a fintech startup, is dedicated to democratizing financial literacy by imparting investment knowledge through personalized large-language modeling (LLM) modules, complete with interactive quizzes for everyday investors. Its mission is to make financial education more accessible and personalized for everyone.

A cloud native company, GoodWhale chose to launch its platform on **Google Cloud** (<https://cloud.google.com/>) due to its intuitive out-of-the-box solutions and flexibility to pick and choose the products they need.

"Being a startup, we didn't want to spend too much time focusing on DevOps. Google Cloud products are pretty intuitive and easy to use so we can pick the products we need, and plug them in. It allows us the flexibility to build up as we go along. We also like

the fact that it is very transparent with pricing models, which helps us better plan and forecast what we need for the future as we continue to grow," says Junxian Lee, Co-founder and Chief Product Officer at GoodWhale.

Prior to implementation, GoodWhale sought the support of CloudMile (<https://mile.cloud/>). "We had discussions with CloudMile about our challenges, and they recommended different approaches we may have overlooked," says Lee. The CloudMile team also provided training opportunities to GoodWhale, focusing on the tools relevant to their business and bringing with them an in-depth understanding of Google Cloud products.

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Building a pedagogy-based platform with deep insights

GoodWhale aims to humanize finance by taking the finance jargon out for the layman. "Some of us used to teach investing and what we found was that a lot of insights comes from the day-to-day interactions with people around topics that are deeply personal," says Lee. "Background narratives are crucial and we need them to align a person's personal values to

their personal finance. That is very difficult to do with technology."

For now, GoodWhale is focusing on getting its users interested in learning more about financial investing by providing courses on its platform, and generating content to tell the stories of companies across different industries. To remain neutral and unbiased, the platform is purely for education and refrains from investment platform recommendations.

"We want to get people to approach investing differently, by looking into the companies they care about first, and then look at their finances. This is different to the typical investor, who may filter the companies they want to invest in purely based on their financial ratios."

On the infrastructure side, simple gamification and interactions are built to engage users on the front end. Meanwhile, the team spends just one to two hours on gen AI content creation in the back-end with Gemini Pro (<https://deepmind.google/technologies/gemini/pro/>) to support its learning and education modules.

Still in its early stages, Lee and his team are constantly experimenting on how they can best use the capabilities of Gemini. "We're now using Gemini Pro to analyze stocks, where we would summarize company stocks, annual reports, and lengthy financial statements, to generate a simple summary," explains Kevin Koo, vice-president of Engineering at GoodWhale. This cuts down the time users would spend scanning through a company

from about seven minutes to just 15 seconds, so they can quickly move on to the next if it's not something they're interested in.

The team is also leveraging data to better understand its users better based on their engagement with the platform. "By understanding what content interests them, we are able to better provide learning materials that will match their interests and skill level," says Koo.

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Maintaining a secure platform on a reliable architecture

One of the biggest appeals for GoodWhale when choosing a cloud platform was in its intuitiveness and simplicity for deployment. "We decided to go with Google Cloud due to its simplicity compared to other cloud providers. This means we get to move our infrastructure spending around, depending on our needs as we scale," says Koo.

Apart from that, Koo explains that leveraging Google Cloud tools that are readily available also made onboarding new team members easier, since most industry practitioners would already be familiar with the platform infrastructure.

To filter companies, GoodWhale leverages Cloud SQL (<https://cloud.google.com/sql>), particularly PostgreSQL and MySQL, and Firestore (<https://cloud.google.com/firestore>) and Datastore (<https://cloud.google.com/datastore/docs/concepts/overview>) for newer projects where the team is still experimenting around with the features that they are building.

To ensure the highest level of data security, GoodWhale leverages Cloud Build (<https://cloud.google.com/build>), a serverless CI/CD platform, and Secret Manager

(<https://cloud.google.com/security/products/secret-manager>)

so that engineers can confidently rotate its keys, knowing that it would be updated across all of its builds. This takes away the complexity of having to manage and update every build individually whenever they rotate keys. With Cloud Build, the team has reduced about 50 hours of time spent on work, per engineer.

For API responses, the team leverages Cloud Memorystore (<https://cloud.google.com/memorystore>) with Redis, serving users data with cache instead of querying the database directly, to speed up responses, This has reduced load time by 200%. The team also taps on BigQuery

(<https://cloud.google.com/bigquery>) and Looker Studio (<https://cloud.google.com/looker-studio>) to analyze user behavior and metrics. This insight gives GoodWhale a better understanding of how users are interacting with the platform, leading to the opportunity to improve the user experience through platform personalization.

As a whole, being on Google Cloud has helped GoodWhale operate on a large scale with a lean team. "We've got a load balancer to deal with traffic spikes, and the Cloud Run

(<https://cloud.google.com/run>) and Cloud Functions (<https://cloud.google.com/functions>) instances are set to automatically scale. This is a game changer for us especially during events like conferences or big consumer events, because we're able to scale up, and then quickly scale back down again," says Koo.



Taking the next leap with Gemini

With a solid infrastructure in place, GoodWhale is set to modernize its applications with generative AI (Gen AI). However, the team is taking one step at a time, experimenting along the way to ensure that they apply the powerful capabilities of Gemini only to areas where it would best serve users.

"Of course, to deliver the best user experience to customers, we have to pick the areas where Gen AI would be best used. There's a long road ahead of us, but I believe that we've already got the tools in place," says Lee.

"We're excited about the potential of Gen AI and what Gemini Pro can do for us, and we want to be at the forefront of this new scene as we build new user experience and consumption habits."

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